The beauty of difference array technique in problem solving

Palak Thapar Follow 2 min read · Mar 13, 2025

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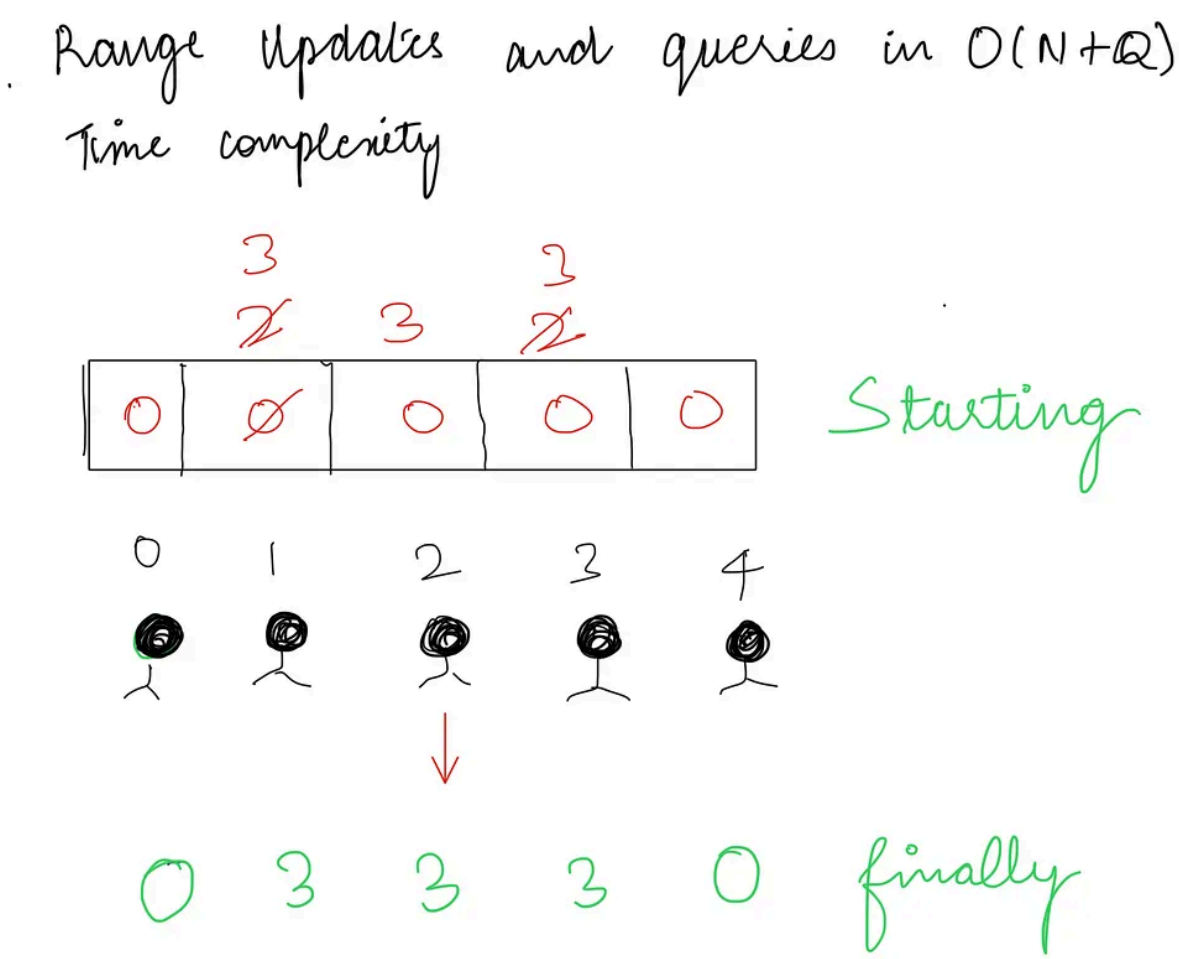
> *Let's say we come across problem statement like the following:*
> *You have an array of students where each index represents points each student has. We get 2D array of queries[L, R, X], where all the students in range L to R get X more points. Then we are asked to find final points of any student.*



Difference array technique

The first idea that we get is, for every query iterate over each element from L to R, we update the index value. But if we look closely, The complexity will go upto O(N*Q) where Q is the size of queries and N is teh size of array. 🥺

Let's learn another approach to do this.

1. Create a difference array of N+1 size named diff.
2. For each query do diff[L]+=X and diff[R+1]-=X.

3. Create a prefix array of N+1 and update each index with pre[i-1]+diff[i]
4. Use this prefix array to return value of points at asked index in O(1)
   This way the total complexity reduces it **O(N+Q), O(Q) for creating the diff array and O(N) for prefix array.** 🤩

Let's do a dry run.

```
N = 6 (Number of students)
Initial Points: [0, 0, 0, 0, 0, 0]
Queries:
1. (1, 3, 2)  → Add 2 points to students in range [1,3]
2. (2, 5, 3)  → Add 3 points to students in range [2,5]
```

Step 1: Apply Difference Array Technique

Initialize `diff` array:

```
Index:    0   1   2   3   4   5
Diff:     0   0   0   0   0   0
```

Query [1, 3, 2]

Add **2** at index `1` → `diff[1] += 2` Subtract **2** at index `4` (`R+1`) → `diff[4] -= 2`

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Updated `diff` array:

```
Index:    0   1   2   3   4   5
Diff:     0   2   0   0  -2   0
```

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- Add **3** at index `2` → `diff[2] += 3`
- Subtract **3** at index `5` (`R+1`) → `diff[5] -= 3`

Updated `diff` array:

```
Index:    0   1   2   3   4   5
Diff:     0   2   3   0  -2  -3
```

Step 2: Compute Prefix Sum to Get Final Points

Now, we compute the **prefix sum** of the `diff` array to get the final values of the student points.

```
Prefix Sum Calculation:
Index:     0   1   2   3   4
Points:    0   2   5   5   3
```

This means the **final points for each student are:**

```
Index:    0   1   2   3   4
Points:   0   2   5   5   3
```

Cool right? 😎
Here is a list of more problems that can be easy peasy lemon squeezy now that you know difference array

https://leetcode.com/problems/zero-array-transformation-i/

https://codeforces.com/contest/276/problem/C

https://leetcode.com/problems/zero-array-transformation-ii/

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